

Morphological Analysis and Finite State Transducer (FST)

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Normalization

- Converting words/tokens in a standard format
- Types:
 - Case-Folding
 - *CAT, Cat, cat* ⇒ *cat* or *CAT*
 - Lemmatization: Finding the root form of the word
 - *is, are, was* ⇒ *be*
 - *cat, cats, goose, geese, fly, flies* ⇒ *cat, cat, goose, goose, fly, fly*
 - Stemming: Chopping off the affixes
 - *is, are, was* ⇒ *is, are, wa*
 - *cat, cats, goose, geese, fly, flies* ⇒ *cat, cat, goos, gees, fli, fli*
 - Others
 - Equivalence class
 - *Normalization* vs *Normalisation*, *color* vs *colour*, *center* vs *centre*
 - *U.S.A.* vs *USA* vs *US*

Morphology

- Field of linguistics that studies the internal structure of words;
 - How they are formed
 - Their relationship to other words in the same language.
- It defines word formation rule from the root word.
- *Morpheme*
 - Smallest linguistic unit that has semantic meaning
 - Can't break down any further without distorting the meaning.

Morphemes

- **Morphemes** can be divided into two part.
 - Stem or the root word: The core meaning bearing unit
 - Affixes: Some part of the surface word that can change the meaning or grammatical functions of the stem:
 - Prefix: “Pre”, “anti”, “in”
 - Suffix: “ed”, “ing”, “s”, “es”, etc.
 - Infix: Inside the stem. Not common in English.
 - Tagalog: *um + hingi* → *humingi* (to ask in Filipino)
 - Circumfixes: Prefix and suffix both
- Cats → Cat + s (One affix)
- Rewrites → Re + write + s (Two affixes)
- Unbelievably → Un + believe + able + ly (Three affixes)

Classes of Morphology

- Inflectional — No changes in the word class
- Derivational — Changes the word class

Inflectional Morphology

- Variations in the form of a word that expresses a grammatical contrast
 - No change in word class
 - Serves grammatical/semantic purposes different than the original form
 - Easy to predict the meaning
 - E.g., “s” or “es” to a noun → defines pluralism
 - OR
 - If you need pluralism → add “s” or “es” to a noun
 - Highly systematic, though some irregularities/exceptions are there
 - Mouse + plural → Mice
 - Can be handled explicitly.

Inflectional Morphology: Noun

- Plural

- Regular:

- “s”

- Most belong to this category

(*bank* → *banks*)

- “es”

- -s at the end
 - -ch at the end
 - -x at the end
 - -y at the end (replace y with ies)

(*class* → *classes*)

(*finch* → *finches*)

(*fox* → *foxes*)

(*fly* → *flies*)

- Irregular

- mouse → mice
 - ox → oxen
 - goose → geese

- Possessive: Apostrophe + -s

- *Children's candies*
 - *Lionesses' kill*

if -s or -z is at the end of plural nouns or names

Inflectional Morphology: Verb

- Regular

- Same affixes mark same functions

- *Walk, push, merge, etc.*

- -s or -es form: *walks, pushes, merges*
- -ing form: *walking, pushing, merging*
- -ed form: *walked, pushed, merged*

- Irregular

- Idiosyncratic forms of inflections

- *Eat, Cut, Catch, Come, Go, etc.*

- -s or -es form: *eats, cuts, catches, comes, goes*
- -ing form: *eating, cutting, catching, coming, going*
- Past form: *ate, cut, caught, came, went*
- -ed form: *eaten, cut, caught, come, gone*

Derivation Morphology

- Combination of a stems with other morphemes changes the class
- Formation of noun from verb/adjective (**nominalization**)
 - Summarize + **ation** → Summarization
 - Trust + **ee** → Trustee
 - Kill + **er** → Killer
 - Fuzzy + **ness** → Fuzziness
- Formation of adjective from noun/verb
 - Computation + **al** → Computational
 - Trust + **able** → Trustable
 - Fear + **less** → Fearless

Morphology Parsing

- Two questions
 - *What is the plural of cat?*
 - *What does cats means?*
- Mapping the surface-level form to the lexical-level form
 - cats → Surface-level
 - cat + N + PL → Lexical-level
- Morphological recognition / analysis: Surface to lexical
- Morphological generation / synthesis: Lexical to surface

Morphology Parsing

Morphological features

Input	Morphological Parsed Output
cats	cat +N +PL
cat	cat +N +SG
cities	city +N +PL
geese	goose +N +PL
goose	(goose +N +SG) or (goose +V)
gooses	goose +V +3SG
merging	merge +V +PRES-PART
caught	(catch +V +PAST-PART) or (catch +V +PAST)

Ambiguity

- Khāyegā → Khā + ye + g + ā
eat 3rd per future male
- Khāyegī → Khā + ye + g + ī
eat 3rd per future female
- Khāongā → Khā + on + g + ā
eat 1st per future male

Issues with Morphology

- Invalid stems

- Going → go + ing 
- Duckling → duckl + ing 

- Some combinations of affixes and stem are not valid

- Doer → Do + er 
- Beer → Be + er 
- Undo → un + do 
- Ungo → un + go 

- Some spellings are not valid

- Boy + s → boys 
- City+ s → Citys 
- Make + ing → makeing 

Morphology Parsing

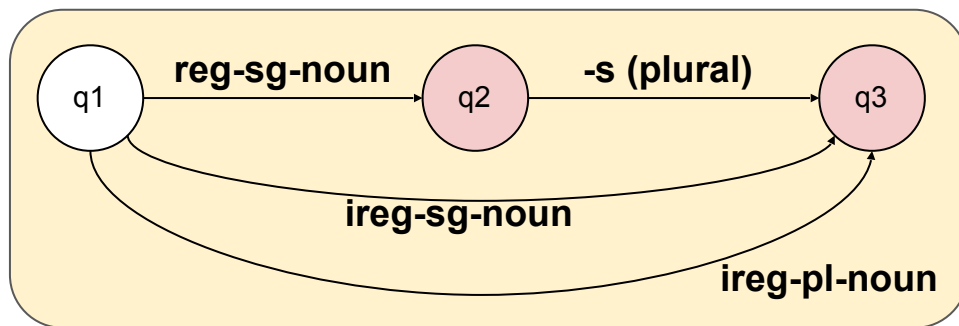
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Morphology Parsing

- Three ingredients for the morphological parser
 - Lexicon
 - List of stems, affixes, and other information (POS tag of stem, etc.)
 - Morphotactics
 - Rules for morpheme ordering, e.g., plural **-s** should follow a noun stem
 - Orthographic rules
 - Rules that defines change in characters, e.g., *city* + *s* → *cities*

FSA and Morphology

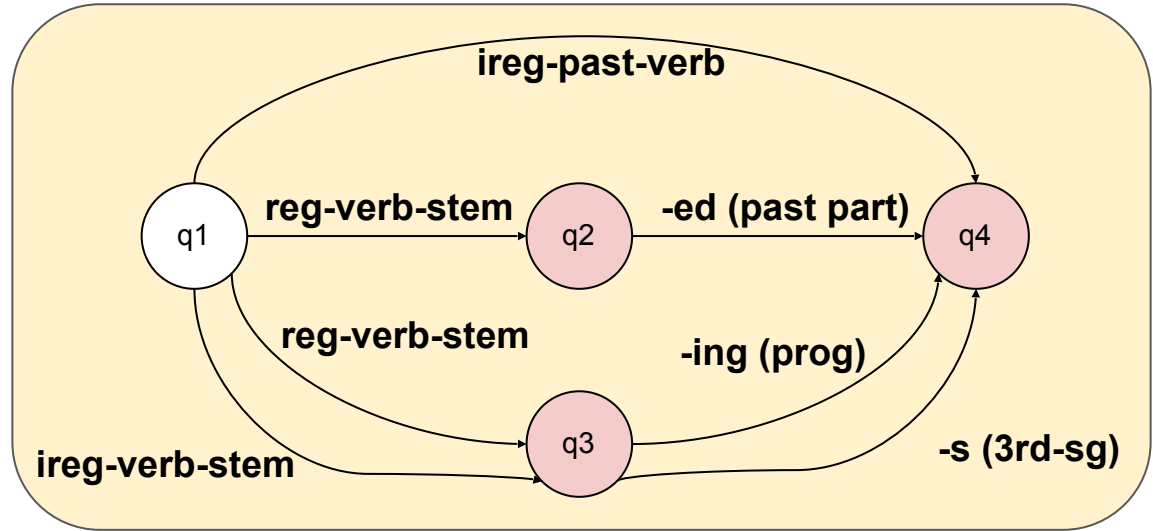
- Recall, FSA are the recognizer/accepter
- We can employ FSA for morphology recognition as well.
- **Ex 1:** Modelling morphotactics for nominal inflectional morphology
 - Regular noun: singular (boy, bank) + plural (boys, banks)
 - Irregular noun: singular (goose, mouse) + plural (geese, mice)



For the simplicity, treating -s and -es plural forms as -s

FSA and Morphology

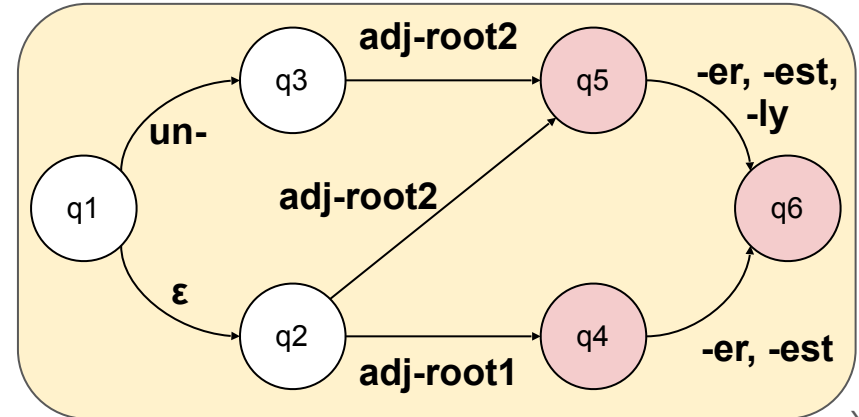
- **Ex 2:** Modelling morphotactics for verbal inflectional morphology
 - Stems:
 - Regular verb
 - Irregular verb
 - Affixes:
 - -ed Past
 - -ed Past Participle
 - -ing Progressive
 - -s 3rd Singular



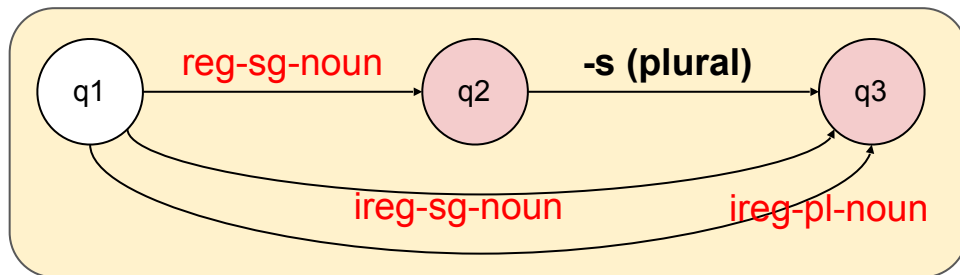
FSA and Morphology

- **Ex 3:** Modelling morphotactics for adjective derivational morphology [Antwork, 1990].
 - List of adjectives
 - big, bigger, biggest, cool, cooler, coolest, coolly, red, redder, reddest, clear, clearer, clearest, clearly, unclear, unclearly, happy, happier, happiest, happily, unhappy, unhappier, unhappiest, unhappily

- Adjectives have
 - Optional prefix (e.g., *un-*)
 - Optional suffix (e.g., *-er*, *-est*, *-ly*)
 - Adj-root1: Can't take *un-* and *-ly* (e.g., big, red)
 - Adj-root2: Can take *un-* and *-ly*

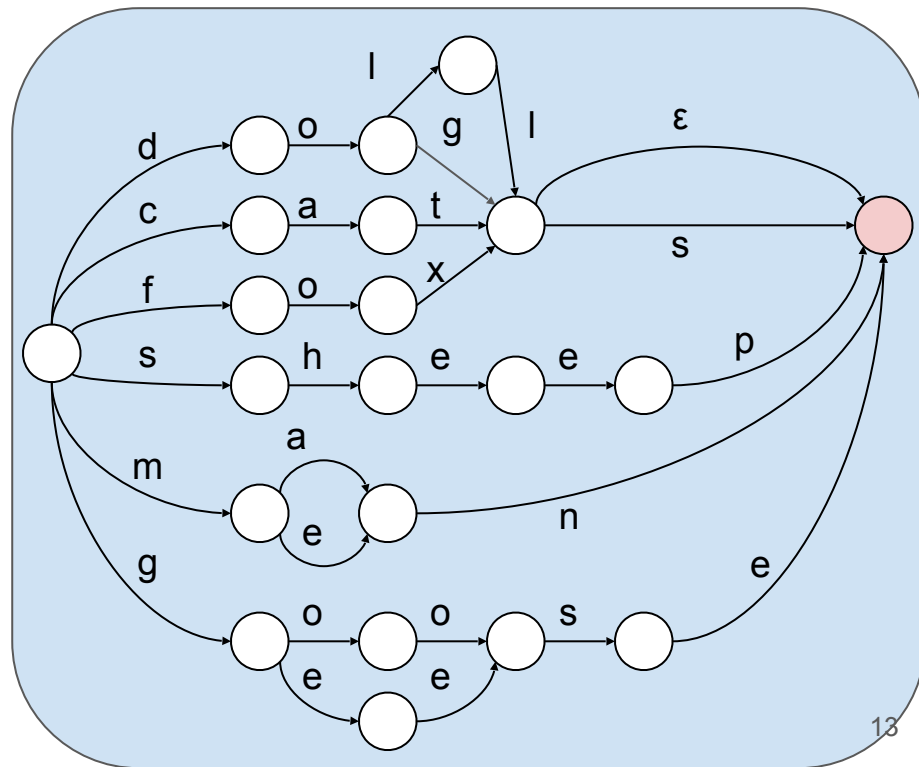


Morphotactics + 'Sub-lexicon'



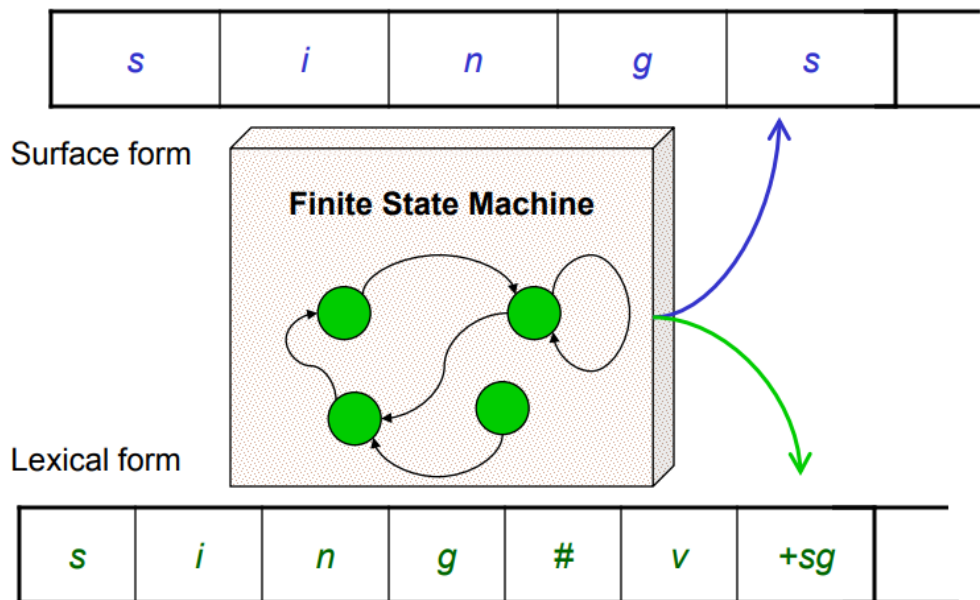
Sub-lexicons:

fox, cat, dog, doll, goose, geese,
man, men, sheep



Morphological Parsing and FST

- Finite State Transducer (FST)
 - Maps between surface-level and lexical-level



What FST can do?

- Acceptor — (DFA)
- Recognizer
 - Given a pair of strings, accepts if they are part of the string-pair language.
- Generator
 - Given a FST, it can generate a pair of output strings
- Translator
 - Given an input string, it can translate it to another language

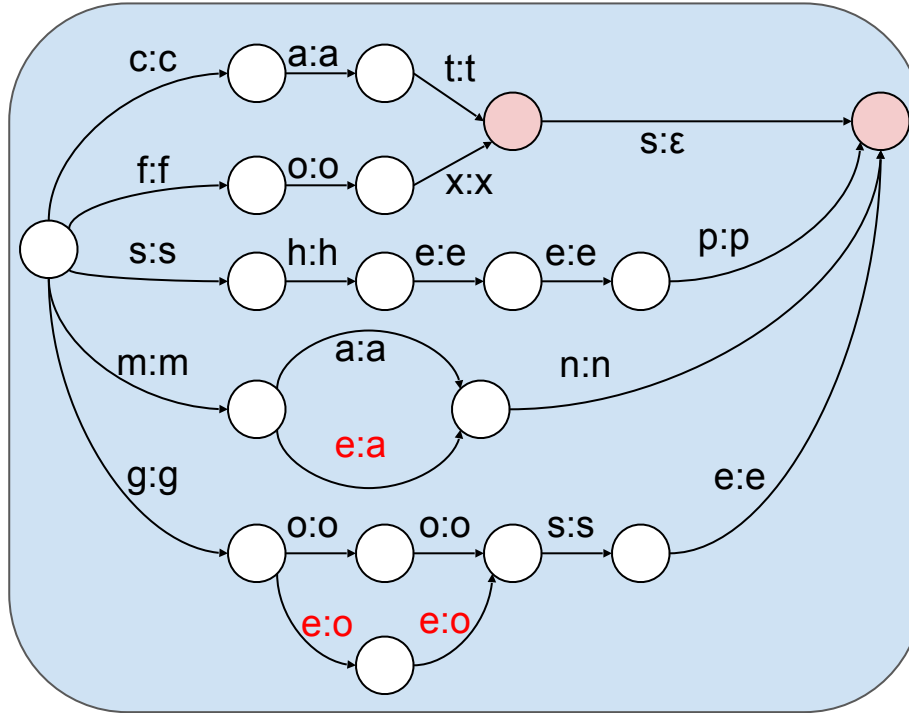
FST

- Extension of FSA that can generate output symbols.
- A 6-tuple $\{Q, \Sigma, \Delta, S, F, \delta\}$
 - Q is the finite set of states
 - Σ is the finite set of input symbols
 - Δ is the finite set of output symbols
 - $S \in Q$ is the start state
 - $F \subseteq Q$ is the set of accepting states
 - δ is the transition function $Q \times \Sigma$ to $Q \times \Delta$

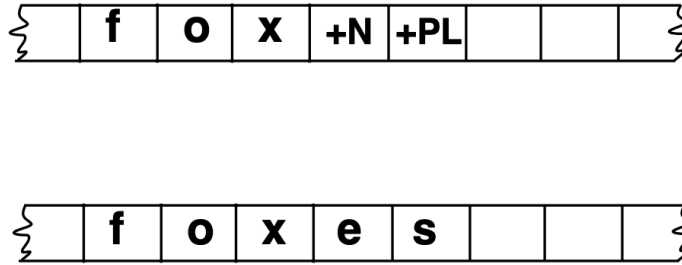
FST: An example

What is this FST doing?

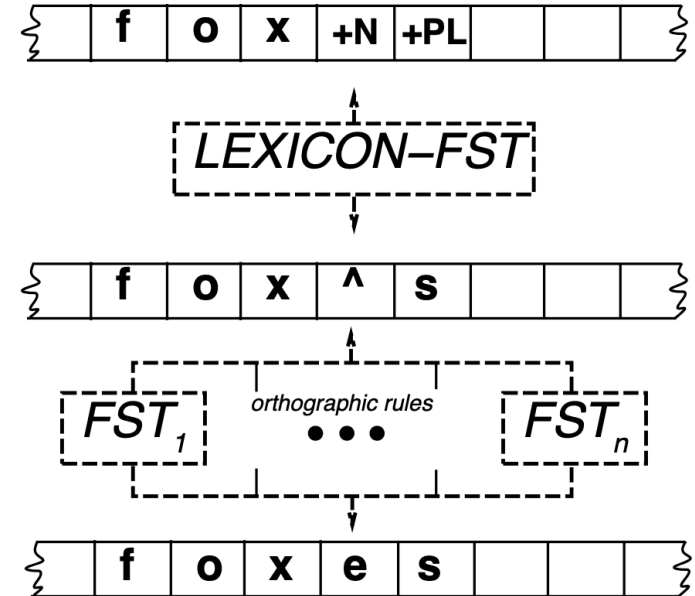
- Returns singular noun



Two-level morphology

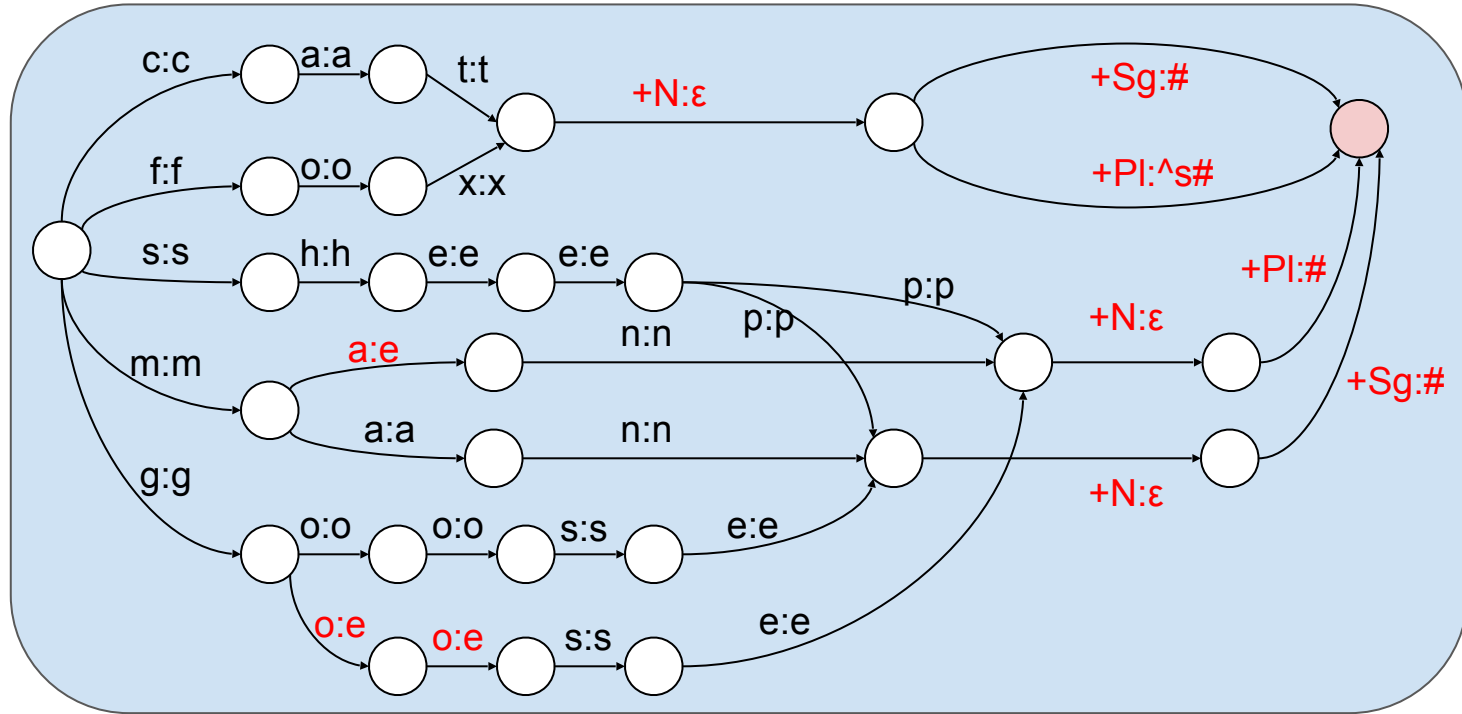


- Lexical-level morphology (lexical and intermediate tapes)
- Surface-level morphology (surface and intermediate tapes)



FST: The lexical example

Lexical → intermediate



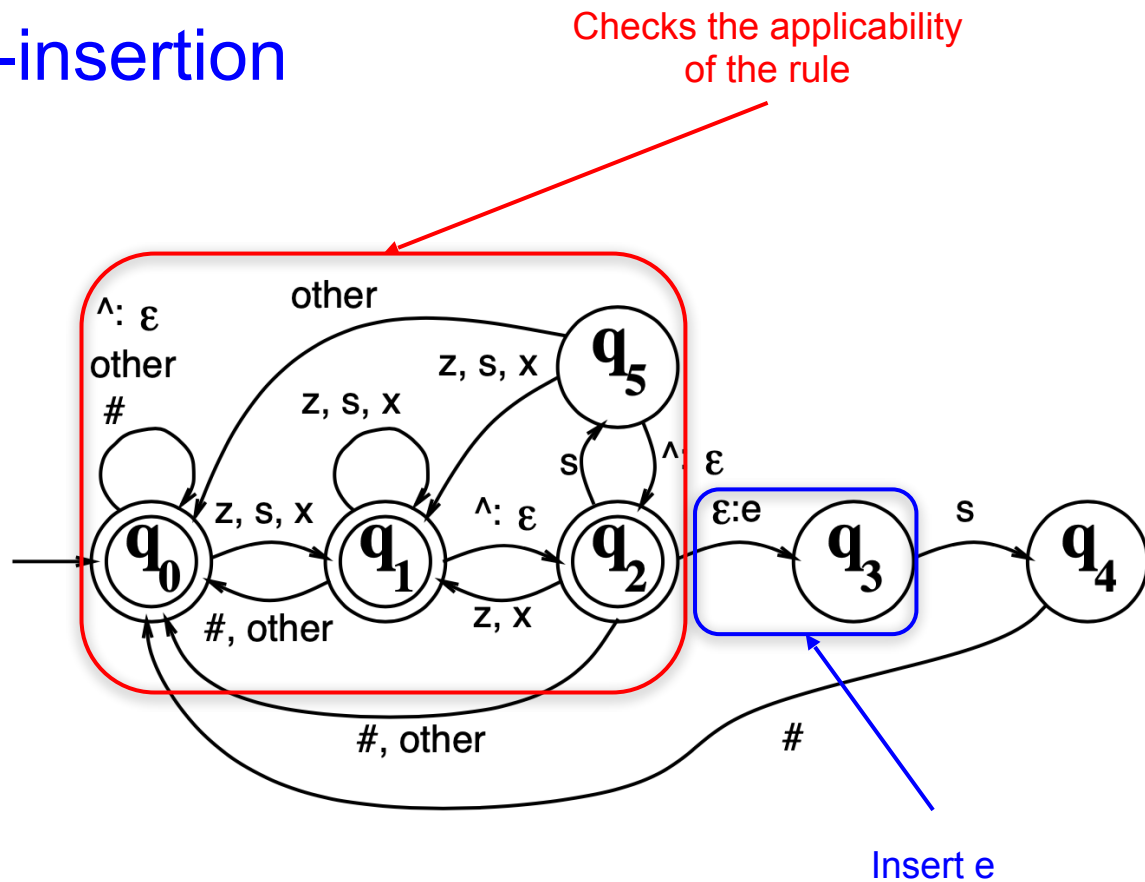
Orthographic (spelling) rules

$$FST_i: \quad \varepsilon \rightarrow e / \left\{ \begin{array}{c} x \\ s \\ z \end{array} \right\}^{\wedge} \text{---} s\#$$

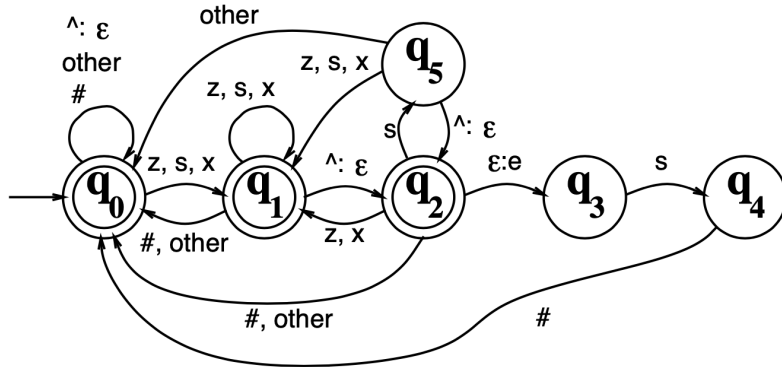
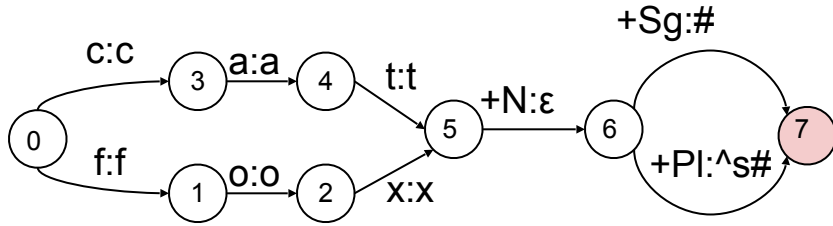
- Spelling changes at the morpheme boundaries

Name	Description of Rule	Example
Consonant doubling	1-letter consonant doubled before <i>-ing/-ed</i>	beg/begging
E deletion	Silent e dropped before <i>-ing</i> and <i>-ed</i>	make/making
E insertion	e added after <i>-s,-z,-x,-ch, -sh</i> before <i>-s</i>	watch/watches
Y replacement	-y changes to <i>-ie</i> before <i>-s, -i</i> before <i>-ed</i>	try/tries
K insertion	verbs ending with <i>vowel + -c</i> add <i>-k</i>	panic/panicked

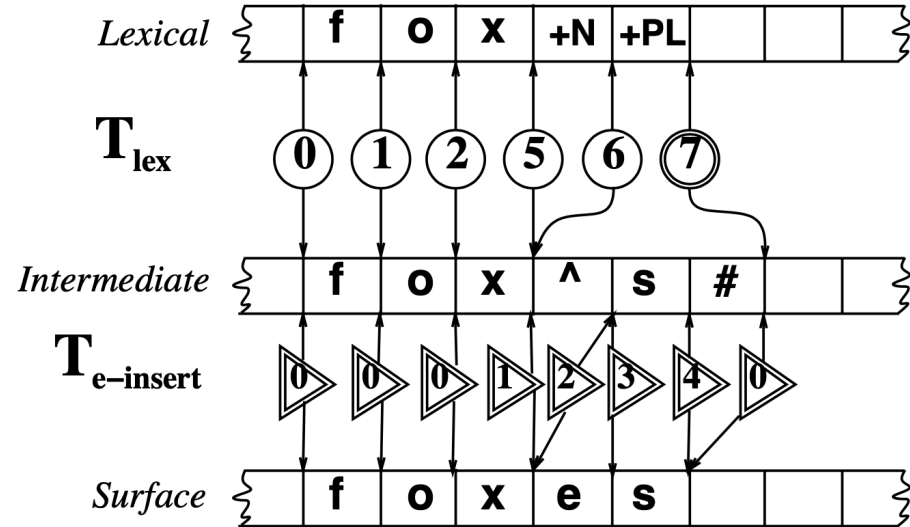
FST for e-insertion



Combine lexical and e-insertion FSTs



FST for e-insertion



Analysis vs Synthesis

- Synthesis (generation) is easier than Analysis (recognition/parsing)
 - Why?
 - Ambiguity
- Synthesis:
 - catch + V + Past $\Rightarrow$ caught
- Analysis
 - caught $\Rightarrow$ catch + V + Past
 - OR
 - caught $\Rightarrow$ catch + V + Past-Part



Disambiguation

- Utilize external evidence, e.g., context
 - May not be available always.
- If no evidence,
 - Output the first solution
 - Output all the solutions
 - Bias the search such that preferred paths are explored

Are lexicon and rules mandatory?

- Lexicon-only morphology
 - Lists all surface level and lexical level pairs
 - Surface $\longleftrightarrow$ Lexical
 - No rules
 - Analysis and Synthesis are easy.
 - Difficult to record all possibilities for any language
- Stemming (Lexicon-free)
 - Set of rules
 - Interested in stems.
 - Don't care about the structure of the word
 - Don't care about the right stem, as long as we get consistent stem.

Porter's Stemmer [Porter, 1980]

- The most popular stemming algorithm
- Applies a series of rules

Original text: India has recorded the biggest single-day spike of 66,999 COVID-19 cases, taking the total number of infections to 23,96,637. However, the recovery rate has gone up to 70.76 per cent, with 16,95,982 people recovering in the country from the highly-contagious disease, government data this morning showed. With the death of 942 patients in the last 24 hours, the country's fatality count rose 47,033, the Union Health Ministry said. India is the third worst-hit country by the pandemic after the United States and Brazil.

Stemming: india **ha record** the biggest single - day spike of 66,999 covid-19 **case** , **take** the total number of **infect** to 23,96,637 . **howev** , the **recoveri** rate **ha** gone up to 70.76 per cent , with 16,95,982 **peopl recov** in the **countri** from the highly - **contagi diseas** , **govern** data **thi morn show** . with the death of 942 **patient** in the last 24 **hour** , the **counti** 's **fatal** count rose 47,033 , the Union Health **Ministri** said . India is the third worst - hit **countri** by the **pandem** after the **unit state** and brazil .

Lemmatization: india **have record** the **big** single - day spike of 66,999 covid-19 **case** , **take** the total number of **infection** to 23,96,637 . however , the recovery rate **have go** up to 70.76 per cent , with 16,95,982 people **recover** in the country from the highly - contagious disease , government data this morning **show** . with the death of 942 **patient** in the last 24 hour , the county 's fatality count **rise** 47,033 , the union health ministry **say** . india **be** the third worst - hit country by the pandemic after the united state and brazil .

Porter's Stemmer [Porter, 1980]

Rule 1.a.

- | | | | |
|---------|------|----------|----------|
| 1. SSES | → SS | caresses | → caress |
| 2. IES | → I | ponies | → poni |
| | | ties | → ti |
| 1. SS | → SS | caress | → caress |
| 2. S | → ε | cats | → cat |

Rule 1.b.

- | | | | |
|--------------|------|-----------|-----------|
| 1. (m>0) EED | → EE | feed | → feed |
| | | agreed | → agree |
| 1. (*v*) ED | → ε | plastered | → plaster |
| | | bled | → bled |
| 1. (*v*) ING | → ε | motoring | → motor |
| | | sing | → sing |

Rule 1.c.

- | | | | |
|------------|-----|-------|---------|
| 1. (*v*) Y | → I | happy | → happi |
| | | Sky | → sky |

If Rule 1b.2 or 1.b.3 is successful, the following is done:

- | | | |
|----------------------------------|-----------------|------------|
| 1. AT -> ATE | conflat(ed) | → conflate |
| 2. BL -> BLE | troubl(ed) | → trouble |
| 3. IZ -> IZE | siz(ed) | → size |
| 4. (*d and not (*L or *S or *Z)) | → single letter | |
| | hopp(ing) | → hop |
| | tann(ed) | → tan |
| | fall(ing) | → fall |
| | hiss(ing) | → hiss |
| | fizz(ed) | → fizz |
| | fail(ing) | → fail |
| | fil(ing) | → file |
1. (m=1 and *o) -> E

Rule 2

- | | | | |
|------------|--------|-------------|-------------|
| 1. ATIONAL | → ATE | relational | → relate |
| 2. TIONAL | → TION | conditional | → condition |
| | | rational | → rational |
| 1. ATOR | → ATE | operator | → operate |

Rule 3

Rule 4

Rule 5

Summary

- Morphological parsing is the process of finding the constituent **morphemes**
- Inflectional and Derivational morphology
- Lexicons, Morphotactics, and Orthographic rules
- Finite State Transducer
- Two-level Morphology
- Porter's Stemmer

Thanks